

Justin Meimar

meimar@ualberta.ca [GitHub](#)

Systems Software Engineer | Edmonton, Alberta, Canada

EDUCATION

University of Alberta - Thesis-Based MSc. Computer Science *Sep 2025 – Aug 2027*

GPA: 3.8 / 4.0

Key Classes: Formal Verification, Advanced Compiler Design, Machine Learning II, Convex Optimization.

University of Alberta - BSc. Spec. Computer Science *Sep 2021 – April 2025*

GPA: 3.7 / 4.0

Key Classes: Algorithms, Compilers, Software Engineering, Operating Systems, Computer Vision & Machine Learning.

EXPERIENCE

Research Assistant | Compiler Design Optimization Lab — Mozilla-sponsored (*Edmonton, Alberta*) *Sep 2025 – Present*

- Designed and implemented AmberMonkey, an AOT code-generation target in SpiderMonkey that reuses Baseline and inline-cache JIT artifacts across processes.

Tensor Compiler Intern | Huawei (*Edmonton, Alberta*) *May – Aug 2025*

- Designed scheduling algorithms using hardware heuristic-guided search to efficiently lower tiled computation graphs to custom matrix accelerators.

Teaching Assistant | University of Alberta (*Edmonton, Alberta*) *Jan 2023 – Dec 2025*

- Head TA for CMPUT 415 Compiler Design (Fall 2024 and 2025), teaching LLVM, MLIR, and practical compiler development.
- TA for CMPUT 429 and 229 Computer Architecture (Winter 2024 and 2023), including RISC-V assembly programming.

Summer Research Intern | Compiler Design Optimization Lab (*Edmonton, Alberta*) *May – Aug 2024*

- Researched copy-and-patch JITs; upgraded compilers to LLVM 18 and added MLIR dialects and lowerings.

Full Stack Intern | Bits in Glass (*Edmonton, Alberta*) *May – Aug 2023*

- Developed an internal web tool in Python and managed deployment on Heroku with Docker.

PROJECTS

Hack GPT Edmonton | 1st Place *July 2023*

- Hackathon - Worked in a team of four to build an interactive information retrieval system for Real Estate listings using Vector-Embeddings and LLM (large language model) APIs.

Home Server | System Administration *June 2023*

- Managed a small server with NGINX to host services over SSH and HTTPs.

Gazprea Compiler | 1st Place *Fall 2022*

- Worked in a team of four to create a tournament winning compiler; focused on type-checking, type-promotion, and control flow implementation using LLVM IR-builder.

STACK

Compiler & Systems / JIT and AOT Compilation - Code Generation - Performance Analysis - Linux

Tools / LLVM - MLIR - Git - CMake - GDB - Valgrind - Docker

Languages / C++ - C - Python - JavaScript - TypeScript - Rust - Bash - SQL